



A Benchmark for Selecting Real-Time Scheduling Approaches for Fleets of Automated Guided Vehicles in Intralogistics

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Abstract In intralogistics, fleets of automated guided vehicles are becoming a key technology offering competitive advantages in performance. Scheduling automated guided vehicle fleets is a complex problem involving the spatial and timely coordination of vehicles and transported cargo. So far, scheduling has been addressed with different traditional approaches as well as machine learning approaches, for example deep (reinforcement) learning models. However, despite the increasing number of approaches, there is no guidance for their selection to optimize fleet efficiency. Hence, decision-makers risk selecting a subpar approach sacrificing efficiency and potentials for competitive advantages. Drawing on the “No Free Lunch Theorem” and using design science, this research contributes with five design requirements, three design principles, and seven design features for a benchmark that ranks approaches for the scheduling of fleets of automated guided vehicles. For the purpose of demonstration and evaluation, the authors instantiated the design knowledge in a benchmark applied in a case study. In terms of implications for practice and theory, the normative knowledge on approach selection can be appropriated to

related problems against the background of proliferating AI-based models.

Keywords Automated guided vehicle fleet scheduling · Benchmark · Decision support systems · Machine learning

1 Introduction

Autonomous locomotion is a field of great interest both in research and everyday life. While self-driving cars are not widely present on public roads yet, automated guided vehicles (AGVs) have been used in industrial automation for several years (Fragapane et al. 2021; de Ryck et al. 2020). In industrial intralogistics, the need to reduce costs and save time increases pressure towards higher operational efficiency, making successful automation using AGVs with best possible transportation schedules a crucial competitive advantage. Recognizing the potentially superior performance, companies have successfully deployed AGVs for transportation in various areas of intralogistics, such as manufacturing lines, warehouses, and container terminals (ten Hompel, 2020; Ullrich and Albrecht 2023; de Ryck et al. 2020).

With flawless operations of AGV fleets remaining imperative to avoid inefficiencies and costly accidents, their scheduling challenges practitioners with a complex optimization problem. Specifically, the planning of AGVs consists of the subproblems of scheduling and routing, where scheduling entails allocating tasks to AGVs over time and routing computes the corresponding trajectories (Fragapane et al. 2021). The engineering of approaches to solve these problems has recently gained momentum due to AI-based methods yielding improved results over the state-of-the-art. This increased interest has led to a stark

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proliferation of scheduling approaches (Aizat et al. 2023, 2024). Traditionally, AGV fleet scheduling has been approached using traditional approaches, such as heuristics for solution approximation (e.g., de Ryck et al. 2020; Xing et al. 2023). With the rise of powerful machine learning (ML) models, deep learning seems to be a natural candidate to challenge these approaches utilizing complex statistics. The models' capability to learn bears performance advantages once more data becomes available. Particularly, deep reinforcement learning (DRL) can be used to take advantage of feedback to dynamically schedule in complex real-world applications in intralogistics (Arques Corrales and Gregori 2020; Hu et al. 2020).

However, referring to the “No Free Lunch Theorem” (Wolpert and Macready 1997), there is no single approach, no silver bullet, that works best in any given context. This is due to the wide variety of use-case-dependent requirements and metrics for approach assessment, which decision makers need to integrate. Furthermore, the need to consider different approaches is emphasized by idiosyncrasies of problem instances, which are individual combinations of problem variables, such as number of AGVs, environment, or specific time restrictions. While research focuses on the development and application of new approaches (Aizat et al. 2023; Du and Ren 2020; François-Lavet et al. 2018), practitioners potentially miss efficiency gains due to a lack of decision support that guides the selection of the best-suited approaches for their problem instances.

The problem of missing guidance can be solved with a problem-specific decision support system (DSS) (e.g., Bartz-Beielstein et al. 2020; Bischl et al. 2016) that prioritizes different approaches for AGV fleet scheduling. However, the lack of DSS available for this inherently complex problem leads to a two-fold problem characterization and motivation.

In response, we contribute the development of design knowledge for a benchmark capable of ranking approaches for AGV fleet scheduling in intralogistics. The benchmark constitutes a decision support tool guiding the decision-makers' scheduling approach selection process. Thus, we raise the following research question: How can a benchmark to rank approaches for the scheduling of AGV fleets in intralogistics be designed and implemented? We understand the benchmark as a DSS that helps decision-makers to select the best-suited approach for AGV fleet scheduling for their use case (Storey et al. 2024). For its design, we derive knowledge from literature reviews and practice interactions and root it in established design requirements for DSSs. The design knowledge is encapsulated in interdependent design requirements (DRs), design principles (DPs), and design features (DFs). In addition, we demonstrate the benchmark through a prototypical implementation and evaluate it within a case study.

The results have practical implications providing guidance to practitioners for selecting from a growing diversity of approaches for AGV fleet scheduling. In terms of theoretical implications, the results contribute to creating a nascent design theory for approach selection against the background of proliferating AI-based models.

The remainder is structured as follows. In Sect. 2, we outline the background on AGV fleet scheduling and DSSs for approach selection. Section 3 covers the design science research (DSR) methodology used. In Sect. 4, we present the design knowledge for the targeted benchmark. We demonstrate and evaluate the design knowledge on the basis of practice interactions and using a prototypical implementation of the benchmark in Sect. 5. Section 6 comprises of a discussion, implications for practice and academia as well as limitations and future work. Lastly, in Sect. 7, we draw a conclusion.

2 Background

2.1 Automated Guided Vehicles

AGVs are unmanned mobile robots that drive following a predefined guidance system under orchestration by a central system, such as a fleet management system (FMS). Historically, AGV guidance systems were fixed installations, such as mechanical tracks or inductive guidance wires, providing physical restriction of the AGVs' paths. Laser and optical guidance systems allow for more flexibility, where the AGVs' paths are restricted digitally (Ullrich and Albrecht 2023). The FMS makes all decisions for the AGV system and dispatches control commands that are executed by the AGVs. Therefore, on-board sensors and compute power of each AGV are minimal (Fragapane et al. 2024).

In contrast, autonomous mobile robots (AMRs) feature a variety of sensors and on-board compute power to allow independent movement without overarching orchestration. They continuously map their environment, locate themselves, and register obstacles, including other robots. Based on this information, AMRs make decentralized decisions in real-time for accepting tasks and navigating beyond the rather fixed routes available to AGVs (Aizat et al. 2024). AMRs are more flexible at the cost of some performance with larger fleet sizes, as a centralized FMS controlling AGVs can optimize decisions on a system level and use smaller safety margins (Silva et al. 2024).

AGVs and AMRs are primarily used in intralogistics contexts, where they automate material flow by transporting cargo. Generally, they are faster, less expensive, and less error-prone compared to human employees, leading to higher throughput (Ullrich and Albrecht 2023). Also, they

are more flexible and efficient compared to fixed transport systems such as conveyor belts. Their flexibility lies in the ability to control each vehicle in the fleet individually to dynamically respond to changing situations, such as the arrival of new cargo or modifications to the floor layout (Wynn et al. 2014). In intralogistics, important application areas of AGVs span industrial fields such as manufacturing, warehousing, and container terminals, and other fields such as hospitals or hotels (Fragapane et al. 2024). For an extensive overview over concepts and applications of AGVs and AMRs in intralogistics see Fragapane et al. (2024) and Ullrich and Albrecht (2023).

In our work, we focus on centrally orchestrated AGVs, though some of our findings may be transferable towards AMRs. Further, we focus on the subset of industrial contexts, where entire facilities can be automated with minimal human interference by using AGVs for cargo transport.

2.2 Scheduling of Automated Guided Vehicles

Several control problems arise when deploying a fleet of AGVs (Singhal et al. 2017). These problems include the scheduling problem and the routing problem of the AGVs, as well as robot communication and control (Fragapane et al. 2021). In general, scheduling is concerned with completing tasks by allocating resources over a period of time, with a goal of optimizing some performance objectives (Pinedo 2022). In the AGV scheduling problem, the resources are AGVs, and the tasks describe the transportation of goods. This is also called the AGV dispatching or the task allocation problem in literature (Zhang et al. 2024). The AGV routing problem is concerned with moving AGVs along their guidance system without collisions, often with the goal of fulfilling a scheduled transportation task. Many AGV use cases require consideration of the scheduling and routing problems in liaison, as allocating AGVs to tasks over time via a scheduling approach directly links to travel times of AGVs, stemming from the routing approach. Therefore, routing performance typically needs to factor into the scheduling approach (Fragapane et al. 2021).

In addition to performance, AGV scheduling needs to adhere to service-level agreements (SLAs) to ensure reliable deadline adherence. Reliability is key for sound business operations, especially if these directly follow items transportation in intralogistics, such as outbound transportation items (Kumawat et al. 2021).

Referring to performance and reliability, different focus areas of research have emerged that differ by problem characteristics of AGV instances in intralogistics (e.g., Vlachos et al. 2024; Shubyn et al. 2023; Wang et al. 2020). The most common characteristics relevant for scheduling are:

1. *Centralized vs. Decentralized*: Decision-makers can schedule the entire AGV fleet centrally, where AGVs do not possess proprietary decision-making capacity. Alternatively, each AGV can make its own decisions decentrally as it is usually the case for AMRs (Fragapane et al. 2021).
2. *Online vs. Offline*: Decision-makers can update schedules online. That is, AGVs are coordinated based on real-time data including, for example, the position of all AGVs and newly incoming cargo they transport. Alternatively, decision-makers can set a schedule for a fixed period of time or number of items the AGVs transport, if the tasks are known in advance (Oyekanlu et al. 2020).
3. *Fixed vs. flexible guidance paths*: AGVs can either drive on fixed routes or, alternatively, use the available free space flexibly. In the latter case, special attention must be given to avoid crashes with objects, such as other AGVs or humans (Bang et al. 2023; Ellithy et al. 2024).

In our research, we focus on completing transport tasks by allocating available robots while optimizing performance criteria, such as completion time, to leverage efficiency (Lu et al. 2022). Related problems, such as inter-robot communication or low-level control, are out of consideration of this research. Depending on the context and use case of an AGV fleet, the scheduling problems are classified by type, such as the (flexible) job shop scheduling problem ((F)JSSP), the (flexible) flow shop scheduling problem, or the automated warehouse scheduling problem. These problems are closely related and have in common that they are NP-hard (Garey et al. 1976; Hoogeveen et al. 1996; Massami and Jin 2011; Xin et al. 2014).

The problem complexity of scheduling AGV fleets as well as the solution potential of ML, especially deep learning, have motivated increasing research interest in recent years (Oyekanlu et al. 2020). Figure 1 shows the increasing attention reflected by a proliferation of publications on AGV fleets including scheduling approaches. The search strings can be found in Appendix A (available online via <http://link.springer.com>).

2.3 Scheduling Approaches for Automated Guided Vehicles

We differentiate approaches to AGV scheduling into *traditional* approaches as well as *ML-based* approaches, where, lately, DRL models seem particularly promising due to their feedback-induced learning capabilities (Hu et al. 2020; Sarkar et al. 2018; Kumar et al. 2011; Zaghoud et al. 2022; Saidi-Mehrabad et al. 2015; Jin and Zhang 2016; Hu et al. 2023). In the following, we delineate

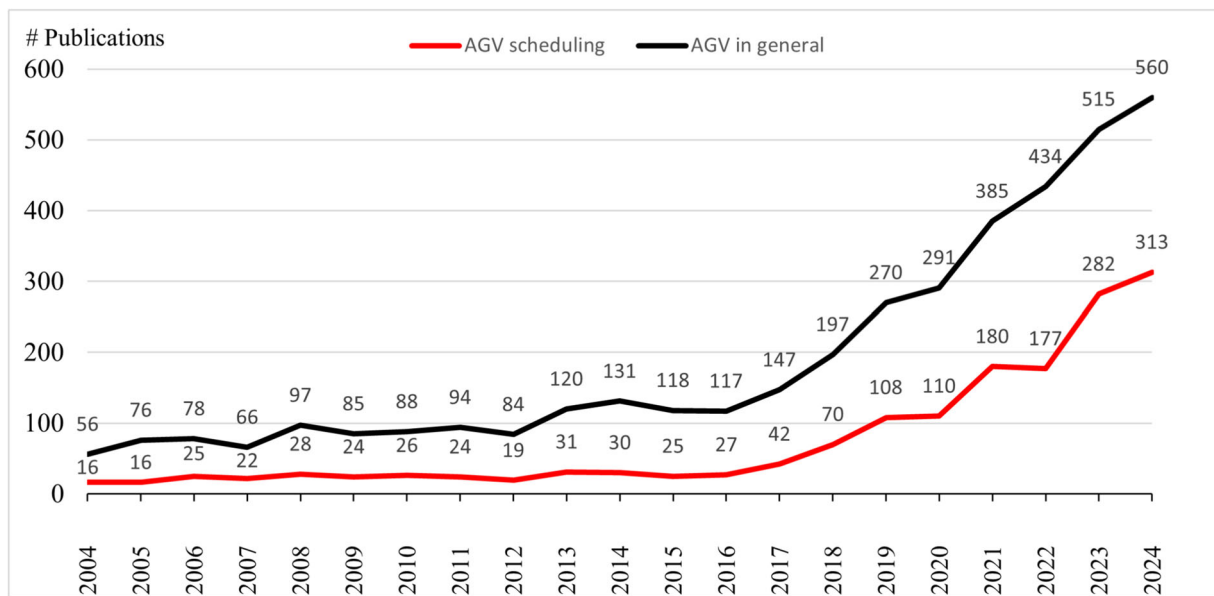


Fig. 1 Development of publications on AGVs in intralogistics and specifically on new approaches for AGV scheduling

traditional approaches from ML-based approaches in more detail including examples.

We use the umbrella term *traditional approaches* to subsume heuristics, symbolic, and non-ML-based optimization algorithms used in literature to schedule AGV fleets.

Heuristic approaches pursue approximation strategies catering to efficiency requirements. Opposed to symbolic algorithms and ML models, heuristics target determining approximate solutions. As the majority of AGV scheduling problems are NP-hard, computing an optimal scheduling solution is infeasible, especially in large industrial AGV systems. While simple heuristics can be implemented quickly, their performance is often subpar. Maximizing system performance with heuristics requires expert knowledge, careful design and extensive testing (de Ryck et al. 2020; Popper and Ruskowski 2022).

Hu et al. (2020) mentions a set of simple heuristics that are applicable to the AGV system, such as first come first served, shortest travel distance first, or nearest vehicle first. The authors used the results achieved by these heuristics as a baseline in their evaluation. Sarkar et al. (2018) proposed a heuristic approach for the scheduling of AGVs in an automated warehouse. Their heuristic was based on using clustering to iteratively find good scheduling decisions. They emphasized fast computation times, even with a large number of tasks, while achieving performance close to that of other proprietary heuristics.

Symbolic algorithms are formal, yet often human-readable and handcrafted representations of problems which result in deterministic, step-wise procedures and seek

comparably systematic solutions to solve AGV scheduling problems.

For example, Li and Kuhl (2017) developed a scheduling algorithm for multiple-load AGVs in a manufacturing context. Their contribution is a rule-based algorithm that regularly determines the next action for each AGV, taking into account the open tasks in the system, the state of load of AGV, and the cost of carrying out each task. The algorithm supports different rule configurations, achieving the overall best results across configurations with a combination of the longest-time-in-system rule and the pickup-or-delivery-en-route rule. An evaluation using two different configurations of the workshop floor shows increased performance compared to other rule-based approaches regarding throughput.

Traditional *optimization* approaches extend the above algorithms by incorporating more sophisticated search strategies that balance exploration and exploitation in complex solution spaces. Among these, meta-heuristics and stochastic optimization have proven particularly effective. They rely on probabilistic rules or iterative improvement mechanisms to escape local optima and approximate solutions efficiently.

Stochastic optimization leverages randomness in the search process to explore large solution spaces effectively (Nea 2022), for example by Monte Carlo simulation scenario generation to determine and evaluate current actions. Meta-heuristics provide general frameworks for problem-solving that can be tailored to specific applications. They typically guide subordinate heuristics through higher-level strategies to improve solution quality iteratively (Gendreau and Potvin 2019). Evolutionary algorithms are a meta-

heuristic known in AGV scheduling. They are inspired by biological evolution, maintain a population of candidate solutions that evolve over generations through selection, crossover, and mutation operators.

For example, Liu et al. (2023) improved performance in a JSSP context by developing the knowledge-guided estimation of distribution algorithm with delivery satisfaction evaluation (KEDAS) using meta-learning. KEDAS is designed as an evolutionary optimization algorithm whose performance is significantly improved by the integration of three knowledge-based strategies, considering dispatching rules, Pareto quality, and delivery satisfaction to guide initialization, offspring generation, and the update of the probability model. The scheduling problem is formulated as a non-linear mixed integer optimization problem and solved by KEDAS.

In contrast, *ML-based approaches* refer to data-driven procedures, where ML algorithms are used to build analytical models with minimal human intervention (Janiesch et al. 2021). Recently, research on the application of ML to AGV scheduling problems has increased and *DRL* models are proliferating for AGV scheduling due to their ability to automatically discover a representation for the learning task.

For example, Arques Corrales and Gregori (2020) solved the integrated scheduling and routing of AGVs in an exemplary warehouse context with two overlapping guidance tracks and four intersections. They proposed a formulation of the problem as a Markov decision process, allowing the application of the proximal policy optimization (PPO) algorithm to learn a scheduling policy. Evaluation in simulation showed that their approach yields valid scheduling results and achieves a throughput close to the theoretical maximum. Hu et al. (2020) proposed a DRL-based approach using the deep q-network algorithm to solve the AGV scheduling problem in a real-world flexible shop floor. Their DRL agent acts as a high-level decision system that is executed for each scheduling decision, selecting a heuristic rule from a predefined set for task assignment, and an AGV to fulfill the selected task. In comparison to the individual heuristics, they find a maximum improvement of 9.8% in makespan, and 10.9% in delay ratio. Further, Zhang et al. (2024) built a DRL system to solve AGV scheduling and routing in a JSSP setting. They employ a digital twin to support improved data collection for their training and decision-making. The routing problem is solved by a deep deterministic policy gradient approach performing collision avoidance on a path generated by an A* approach. A dueling double deep Q network approach solves the scheduling problem by gathering control data from the digital twin and the routing agent. This DRL-based scheduling model achieves improved performance compared to other state-of-the-art methods

with regard to tardiness, stability and energy consumption. Recently, Li et al. (2025a) applied DRL to optimize scheduling for total tardiness in a FJSSP context. They chose DRL due to its capabilities of handling dynamic events in real-time. They based their approach on multi actor reinforcement learning, contributing key concepts for a tardiness-optimizing reward, and a novel encoding of the action space to improve learning results. In an evaluation against other approaches from both the *traditional* and the *ML-based* approach classes, they found performance improvements exceeding 10% in most cases.

As argued above, in the context of ML *DRL* models seem particularly promising due to their ability to acquire information on patterns from data and iteratively improve decisions by maximizing cumulative rewards through agent-environment interactions. In this way, they learn scheduling policies by simulating many scheduling instances and inferring good scheduling decisions. The training process is computationally expensive, as it requires large amounts of data and updates of the underlying neural networks. However, once trained, a DRL-based scheduling approach only requires relatively computationally inexpensive forward passes of the neural networks. To learn from data, expensive pre-processing is required to collect training data and train the scheduling policies (de Ryck et al. 2020; Arques Corrales and Gregori 2020; Hu et al. 2020; Qin et al. 2023; Popper and Ruskowski 2022).

2.4 Decision Support Systems for Approach Selection

DSSs focus on helping users in their decision-making tasks. DSS can be distinguished from decision-making, decision-collaboration, and decision-innovation systems by their supporting role that does not intervene in the decision-making process and leaves the responsibility to the user (Storey et al. 2024). A focus area of DSSs targets selecting best-suited approaches from a pool of alternatives (Rice 1976). This selection is contingent on a variety of factors including for example problem specificity and application scenario (Wolpert and Macready 1997). With an ever-increasing assortment of computational problems and solutions, especially in times of vastly improving computation technology and the entailed rise of ML, the need to select the best-suited approach to solve a given problem has grown (Lee and Shin 2020). DSSs have been shown to outperform experts in approach selection problems and are even more helpful to novice users (Luo 2016).

There are two main methods for approach selection: online and offline DSSs. The former aims to support selecting an approach from an available set of parallel running approaches at runtime. That is, a DSS assesses which approach is suited best for processing or analysis of each individual data point (e.g., Silva et al. 2019). Opposed

to this, the latter is used after several alternative approaches have been applied to a representative data set. For example, benchmarks use pre-defined criteria to assign each approach with a score based on its performance (de Ryck et al. 2020; Heiden et al. 2021). Typically, the approach reaching the best benchmarking score is then chosen as the best-suited approach for application during runtime.

In this research, we focus on identifying design knowledge for offline benchmarks. Specifically, several scheduling approaches are applied centrally for the entire fleet, coordinating the time performance and task allocation of all single AGVs in combination. The benchmark can be used to prioritize the approaches by their performance in a given problem instance that is characterized, for example, by the number of AGVs involved or the number of items to be transported.

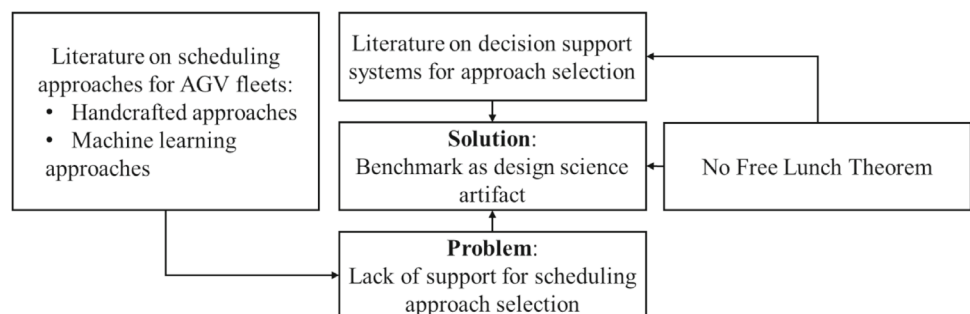
3 Methodology

With our research, we contribute to DSSs research regarding the choice of scheduling approaches for AGV fleets. To this end, we relate our contribution to the respective literature bodies. On the one hand, there is the diverse and proliferating literature on scheduling approaches. On the other hand, we refer to literature on DSSs for approach selection. Our considerations draw on the “No Free Lunch Theorem” stating that there is no single best approach for all instances of a given problem (Wolpert and Macready 1997). This notion of *no size fits all* motivates creating a selection process to determine the best-suited approach for each individual problem instance (e.g., Schneider et al. 2023). In the case of AGV scheduling, problem instances may vary, for example, due to different numbers or kinds of AGVs, different layouts, or individual time constraints. Figure 2 shows the relation (arrows) of related literature bodies and theory (boxes) in the context of our problem in focus and the envisaged solution. That is, the literature body on AGV approaches informed our problem statement, while the theorem and literature on DSSs informed our solution design.

In line with related research (e.g., Ballandies et al. 2025; Schütte et al. 2022), we draw on DSR to design an artifact at the interface of human-computer interaction and consider performance factors as well as user preferences to derive and instantiate the targeted benchmark. DSR focuses on creating and evaluating innovative artifacts, such as models, methods, and systems, to solve identified problems. It emphasizes the practical application of knowledge to develop solutions that improve organizational processes and decision-making (Hevner et al. 2004). In DSR, DRs specify the necessary conditions and functionalities that an artifact must fulfill to address a specific problem. DPs represent prescriptive design knowledge that inform the creation of artifacts, ensuring they are effective, efficient, and user-friendly (Gregor et al. 2020). DFs are the specific characteristics or attributes of an artifact that embody these requirements and principles, contributing to its overall functionality and usability (Meth et al. 2015). To this end, we selected DSR as we target the derivation of actionable, constructivist knowledge that can be reused for practical implementations following common practice in information systems research (e.g., Schütte et al. 2022; Arnott and Pervan 2014). We follow the DSR guidelines established by (Peffer et al. 2007), which comprise six steps. Figure 3 summarizes our research approach, where we combine demonstration and evaluation into one step, so that we have five steps overall.

Define problem and motivate. We started with developing a general, socio-technical understanding of the focused topic including the current state of research and the technical language used in the field. To this end, we engaged with literature and used prior practical experience of one of the authors’ with scheduling AGV fleets in intralogistics. As outlined in Sect. 1, we find a two-fold problem characterization and motivation. On the one hand, we motivate our research with a problematization approach that is due to the complex nature of AGV fleet scheduling and the diversity of solution approaches. On the other hand, we identify a research gap concerning DSSs for model selection to solve the characterized problem. Furthermore, we used the insights from our engagement with literature to identify relevant keyword combinations preparing a

Fig. 2 Relation of literature and theory to problem and solution



	First design cycle	Second design cycle
Define problem and motivate (socio-technical problem understanding)	• Unstructured literature review • Prior knowledge from practical experience	
Define objectives of a solution (derivation of the design knowledge)	• Structured literature review ($n = 25$) • Creation of concept matrix	• Further elicitation of DPs and DFs from expert interviews ($n = 5$)
Design and development (prototypical implementation)	• Identification of DSSDRs • Initial formulation of DRs, DPs, and DFs	• Refinement and formalization of the design knowledge • Prototypical implementation of a benchmark
Demonstration and evaluation (application of implementation)	• Qualitative evaluation within expert interviews ($n = 5$)	• Qualitative evaluation with expert interviews ($n = 4$) • Instantiation and application of the benchmark
Communication (critical reflection)		• Discussion, implications for practice and theory • Limitations • Starting points for future research

Fig. 3 Research approach

structured literature review. Additionally, relevant databases containing literature on the research topic were identified in this step.

Define objectives of a solution. In the first design cycle, we conducted a structured literature review (vom Brocke et al. 2015). Informed by prior engagement with the topic, we created the following search string: *OR* (“AGV”, “autonomous guided vehicle”) *AND OR* (“scheduling”, “dispatching”, “task allocation”) *AND* “learning”. We used the databases *ACM Digital Library*, *IEEE Xplore*, *ScienceDirect*, and *Scopus*. The choice of databases is due to their broad coverage of high-quality outlets encompassing literature on information systems, computer science, and logistics. The initial search resulted in 160 publications. After removing duplicates, 96 unique publications remained. Table 1 shows our literature selection process, where the only exclusion criterion was a deviation from our research focus. Overall, we used 25 publications. To conceptualize the first design knowledge, we created a concept matrix structuring the knowledge derivation process (Webster and Watson 2002). We provide the full concept matrix in Online Appendix B.

In the second design cycle, we conducted interviews with experts from practice to gather additional insight from business practice as well as validate and evaluate our initial findings. The experts were acquired through work-related contacts from one of the authors. We selected experts from companies in the intralogistics automation and research industry. We required the experts to have active

development experience in both the fields of intralogistics and AGV fleets. All experts were working on AGV FMSs at the time of the interviews. In their work, they had direct insight into the relevance and requirements of AGV scheduling. Overall, five experts participated. See Table 2 for details on the experts’ experience and occupation. Three experts worked as software developers, developing and implementing approaches for a FMS. One expert worked as a robotics researcher, focusing on managing an AGV fleet precisely at high velocities. One expert worked as a technical lead on an AGV project, where they worked directly with implementing an FMS, while also regularly communicating with management, gaining additional insights into practical requirements of AGV systems in intralogistics.

Before starting the interviews, the experts were introduced to the general goal of our work and the details of the interview process. The interviews were then structured into three parts. First, the experts were asked about their background and experience in the field of AGV fleets and intralogistics. Second, the experts were asked openly about design aspects and specifically metrics they consider relevant in the benchmark to be designed. To avoid bias, the experts were not presented with any intermediate results. Third, the experts were presented with the design knowledge derived from literature and asked for an evaluation (see “demonstration and evaluation”, first design cycle). The interviews were conducted in English. We transcribed the interviews with the consent of the experts. We analyzed

Table 1 Literature selection procedure

Selection measure	# Resulting publications
<i>Database screening</i>	
Initial results after searching all databases.	160
<i>Remove duplicates</i>	
Remove all duplicates.	96
<i>Coarse-granular selection</i>	
Remove publications where title and abstract show deviation from research focus on AGV fleet scheduling.	74
Remove publications where title and abstract disqualify publication in some other way (e.g., not in German or English).	47
<i>Fine-granular selection</i>	
Remove publications where full text shows deviation from research focus on AGV fleet scheduling.	45
Remove publications where full text shows deviation from research focus by increased focus on multi-objective optimization or incompatible AGV use case.	25

Table 2 Experience of the experts in the fields of AGVs and intralogistics

#	Experience with AGVs	Experience in intralogistics	Role
E1	1 year	2 years	Software developer
E2	< 1 year	11 years	Software developer
E3	2 years	2 years	Software developer
E4	4.5 years	4.5 years	Robotics researcher
E5	7 years	7 years	Technical lead
Average	2.92 years	5.3 years	

the interviews following the guidelines by Mayring (2014). Two researchers analyzed the interview transcripts independently and discussed discrepancies. The Cohen's Kappa for inter-coder reliability was 0.82.

Design and development. In the first design cycle, we identified DSS-specific design requirements (DSSDRs) that we adapted from literature. Also, we formulated initial concrete DRs, DPs, and DFs for the targeted benchmark. In the second design cycle, we refined the derived design knowledge and formalized logical components, such as metrics, mathematically. Further, we prototypically implemented the benchmark in Python 3.11.4.

Demonstration and evaluation. In the first design cycle, we evaluated the existing design knowledge within the expert interviews (cf. details from “define objectives of a solution”, second design cycle). In the second design cycle, the demonstration and evaluation consisted of two parts combining a naturalistic with an artificial evaluation (Venable et al. 2016).

In terms of the naturalistic evaluation, we interviewed four additional experts from practice (see Table 4). We demonstrated the DPs and asked the experts for their evaluation, specifically with regards to solving the identified problem. The interviews were semi-structured. Additionally, we asked the experts to rank statements from

Iivari et al. (2021) on *accessibility, importance, novelty and insightfulness, actability and appropriate guidance* as well as *effectiveness* in practice. Regarding the artificial evaluation, we instantiated the design knowledge to a benchmark and applied it in a case study of an AGV fleet to be scheduled. Using the benchmark, we determined the best-suited approach for scheduling an AGV fleet in four scenarios in a 2×2 factorial design. We used five alternative scheduling approaches: a random approach, two traditional approaches, and two DRL-based approaches as these constitute the state-of-the-art in ML-based solutions for AGV scheduling. After application in a case study, we evaluated the approaches using the instantiated benchmark.

Communication. The results are communicated including a discussion, implications for practice and theory, limitations as well as starting points for future research.

4 Design Requirements, Design Principles, and Design Features for a Benchmark to Rank Scheduling Approaches for AGV fleets

We formulate the identified design knowledge in the form of design requirements (DRs), design principles (DPs), and design features (DFs). To embed the design knowledge into

established research, we build on Meth et al. (2015), who formulate three DSS-specific design requirements (DSSDRs) that have already been adopted in other DSR studies (e.g., Zschech et al. 2020) as they constitute a generic foundation for knowledge creation in the context of decision support. These DSSDRs capture the requirements of human decision makers using a DSS. On this basis, we formulate DRs and DPs using the results from our literature research and expert interviews. The DPs have generic and prescriptive character enhancing their reusability in related contexts. Subsequently, we formulate concrete DFs to translate the DPs towards a prototypical implementation. These DRs, DPs, and DFs represent the final configuration after the second design cycle. Figure 4 shows the identified design knowledge and also makes the relations between the different elements apparent.

4.1 Adoption of Decision Support Systems Design Requirements

We interpret and adopt the DSSDRs of Meth et al. (2015) as follows:

DSSDR1: Increase decision quality. Using DSSs, human decision makers target to maximize their decision quality. Regarding automated guided vehicles (AGVs) scheduling, decision quality relates to choosing the best-suited scheduling approach (e.g., (Xing et al. 2023; Aizat et al. 2023)). Decision quality is measured and compared among competing options, informing the decision.

DSSDR2: Minimize system restrictiveness. Using DSSs, human decision makers aim at maintaining control over the decision process. In this vein, the benchmark needs to guide decision making without compromising the decision-making capacities due to system-specific restrictions, such

as compatibility to only a limited selection of AGV fleet scheduling approaches.

DSSDR3: Reduce cognitive effort. Using DSSs, human decision makers seek to keep the cognitive effort required to make a decision low. This is due to the multitude of decisions necessary in diverse business context and the cognitive limitations of humans. Consequently, the benchmark design should provide decision-relevant information in a manner that is easy to interpret.

4.2 Identification of Design Requirements

Based on our literature search, the interview study and the DSSDRs, we derived the following concrete DRs as necessary conditions and functionalities that a DSS, viz. the benchmark, must fulfill to address the problem of AGV fleet scheduling approach selection:

DR1: Consider throughput. In intralogistics, companies meet increasing pressure towards efficiency. E3 recognized that: “Number one is throughput, that is what everyone seems to talk about.” An optimal schedule for AGV fleets minimizes the makespan (i.e. the time to complete all tasks) and maximizes the throughput (i.e. the amount of cargo transported per time). Throughput allows for a more straight-forward comparison compared to makespan, as it normalizes data to a fixed time frame. The benchmark is required to assess different approaches in terms of throughput to improve overall decision quality in the selection of the best-suited approach (DSSDR1).

DR2: Consider responsiveness. The benchmark needs to measure how fast an approach can issue a schedule based on a given situation of the AGV fleet. The importance of this responsiveness is due to the non-deterministic nature of AGV fleet scheduling, where the handling of single

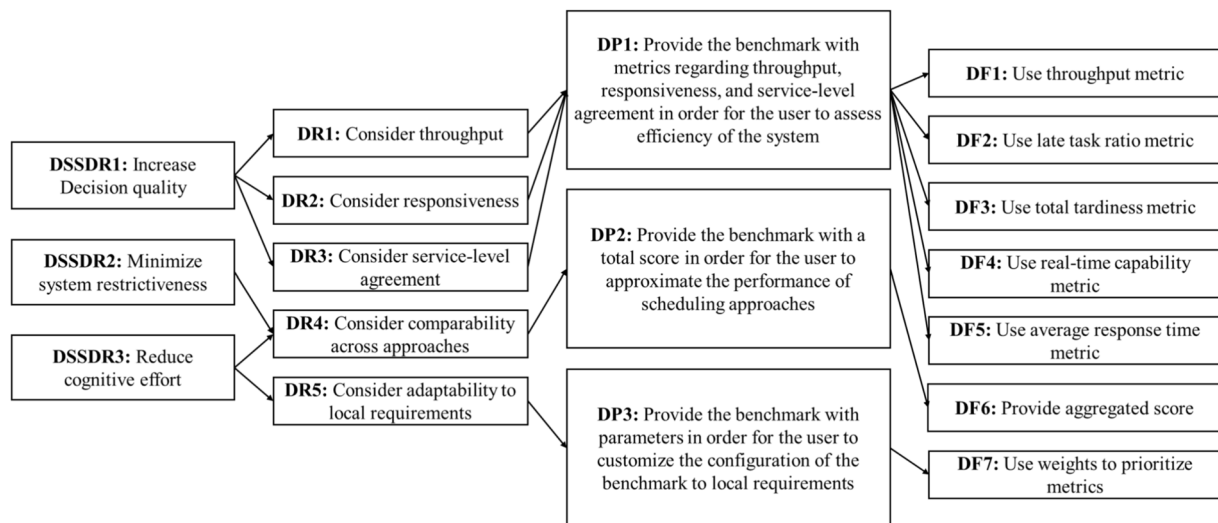


Fig. 4 Design knowledge: design requirements, design principles, and design features

AGVs in a fleet has to be dynamically adjustable to given situations. E5 reinforced this importance, stating: “[...] you have [the problem of] how responsive the AGV system is to a request from, for example, a warehouse. How fast you can [...] drive to a pallet to pick a load.” In this regard, the benchmark needs to consider the real-time capability by considering the effort of an AGV scheduling approach required to compute the schedule. Low responsiveness negatively impacts decision quality (DSSDR1), since the AGV fleet cannot be scheduled as dynamically. Scheduling mistakes that are due to low responsiveness to updates to the schedule could, for example, lead to accidents of AGVs.

DR3: Consider service-level agreements (SLAs). The benchmark is required to maintain the SLAs of intralogistics in terms of completion time. High throughput alone does not guarantee timely completion. In this regard, E2 added: “Thinking as a customer, my primary goal [...] would be [that] whatever work I am getting has [...] to be finished in a timely manner”. Timeliness measurements refer to how much an approach delays cargo transportation, which may lead to breaking SLAs. Delay can result, for example, when an approach issues a schedule that is unrealistic in its execution by the AGV fleet. Referring to DSSDR1, decision quality depends on timely task completion meeting competitive efficiency constraints. The benchmark needs to prefer scheduling approaches that introduce the least delay.

DR4: Consider comparability across approaches. The variety of measures to assess different approaches for AGV scheduling hinders comparability. Integrating various measures has implications on decision-maker’s cognitive load (Sweller 2010). To reduce the cognitive effort (DSSDR3), the benchmark is required to facilitate comparability among approaches based on the variety of metrics. Comparability mechanisms need to be generally understandable and independent of the system the benchmark is deployed to (DSSDR2).

DR5: Consider adaptability to local requirements. To address the manifold use cases and problem instances of AGV fleet scheduling, the benchmark needs to be flexible and minimize restrictions (DSSDR2). This allows it to remain compatible with AGV use cases in general. The need for such flexibility and adaptability was emphasized by E5, “because the project or the field, where we do this automation, is very different from one to the other.” At the same time, the design of broad use case compatibility cannot sacrifice result quality in terms of compromised prioritization of scheduling approaches. A compatible design also reduces cognitive effort (DSSDR3) allowing the decision maker to apply the same benchmark to various specific problem instances without the need to adjust to

instance-specific idiosyncrasies regarding the assessment of different scheduling approaches.

4.3 Identification of Design Principles

In an iterative process as outlined above, we formulated multiple versions of our DPs with the following representing our final configuration. They can be applied in design situations where implementers require guidance to design a DSS for AGV fleet scheduling:

DP1: Provide the benchmark with metrics regarding throughput, responsiveness, and service level agreements (SLA) in order for the user to assess efficiency of the system. The benchmark must incorporate essential metrics that allow for maximum efficiency while maintaining the reliability of AGV fleet scheduling approaches. To ensure users can make well-informed decisions when selecting the most appropriate scheduling approach, these metrics must be designed comprehensively. Building on DR1-3, the benchmark should meet three requirements: throughput, to measure the volume of cargo transported per time unit; responsiveness, to evaluate how quickly and effectively the system can react to real-time changes in the fleet’s environment; and adherence to SLAs, to ensure that schedules meet required deadlines without causing delays. By providing a holistic evaluation of these metrics, the benchmark enables users to select the approach that best balances performance, reliability, and adaptability.

DP2: Provide the benchmark with a total score in order for the user to approximate the performance of scheduling approaches. The benchmark needs to reduce the cognitive effort of its users. As outlined above, DP1 covers different metrics that need to be considered to select the most suitable scheduling approach. These metrics encompass complex, interrelated decision logic. For example, response time directly relates to maximum throughput, where delays can lower the number of transported items in a given period of time. Decision-makers require an abstraction from this complexity to facilitate access to the result interpretation, ranking, and selection of an AGV fleet scheduling approach. Besides this abstraction, the individual metrics still need to be accessible. That is, the total yet only indicative score provides overall comparison among AGV fleet scheduling approaches, a drill-through to the metrics of the approaches enables a more detailed comparison.

DP3: Provide the benchmark with parameters in order for the user to customize the configuration of the benchmark to local requirements. The benchmark needs to be adjustable to individual use cases of AGV fleet scheduling. Demands on AGV fleets can change rapidly when companies change their preferences and, for example, decide to prioritize physical soundness of AGV robots at the cost of throughput. Particularly, customization refers to enabling

decision-makers to configure the assessment of AGV fleet scheduling approaches to individual problem instances. This configuration has to directly address the calculation of efficiency-related metrics (cf. DP1). Customization of selecting AGV scheduling approaches acknowledges that there is no “one size fits all” approach. It increases decision quality, since decision-makers do not have to compromise their selection with generic assessment priorities that have to be valid across situational changes, such as changes in demands on AGV fleets. Moreover, customization to use cases means that the selection of AGV fleet scheduling approaches has to be adaptable to different scenarios, where AGV fleets are deployed. Maintaining the same decision logic for selection reduces cognitive effort as decision-makers do not have to adapt.

4.4 Identification of Design Features

In the following, we further project the DPs into concrete DFs. DFs facilitate the implementation of an instantiated artifact by providing concrete options for a design specification that embody the requirements and principles. We base the mathematical formulation of our DFs on a generalized mathematical notation for AGV scheduling problems inspired by Pinedo (2022) and Smith and Becker (1997). The mathematical scheduling notation is presented in Table 3.

DF1-DF5 define concrete, efficiency-related metrics (cf. DP1). The selected metrics originate from the structured literature review and interviews with practitioners. DF6 comprises the aggregation of these metrics to a single score

to facilitate result interpretation (DP2). Referring to DP3, DF7 enables the individualization to specific use cases based on configurable weights to prioritize the metrics. The DFs build the basis for the prototypical implementation used for the purpose of demonstration and evaluation (cf. Sect. 5).

DF1: Use throughput metric. We opted to use the throughput metric in our benchmark, as it contains implicit normalization to a fixed time frame, for example of one hour, as different from comparable metrics such as makespan (Durst et al. 2023; Zhang et al. 2022). We calculate the throughput following Eq. 1.

$$P = n / \frac{C_{max}}{3600} \left\lceil \frac{tasks}{h} \right\rceil \quad (1)$$

DF2: Use late task ratio metric. The late task ratio refers to the spread of timeliness across all tasks. It indicates the percentage of tasks that are not completed within their deadline. We calculate the late task ratio as defined in Eq. 2.

$$T_r = \left(\sum_{i=1}^n \begin{cases} 1 & \text{if } l_i > 0 \\ 0 & \text{if } l_i \leq 0 \end{cases} \right) / n \in [0; 1] \quad (2)$$

For example, if 1% of tasks are late, then they could equally be late by one second or one hour. We therefore include DF3 focusing on the delay of tasks.

DF3: Use total tardiness metric. Total tardiness captures the overall timeliness of the system by aggregating the delay of each task. It does not indicate how this tardiness is spread across tasks. For example, a total tardiness of 3600 s could either mean 3600 tasks that are one second late, or

Table 3 Mathematical scheduling notation

Symbol/formula	Description
n	Number of tasks
$T = \{T_1, \dots, T_n\}$	Set of tasks
r_i	The requirements of task T_i , abstracting all problem-specific requirements
t_i	The time when task T_i becomes available
d_i	The deadline of task T_i
p_i	The estimated processing time of task T_i
$S = \{A_1, \dots, A_n\}$	The computed schedule, as a set of assignments
a_i	The assignment information of assignment A_i for task T_i , this abstracts all problem-specific assignment information
s_i	The scheduled start time of task T_i
s_i^*	The time an AGV responds to task T_i
p_i^*	The actual processing time of task T_i
e_i	The computation time for scheduling task T_i
$c_i = s_i + p_i^*$	The completion time of task T_i
$h_i = s_i^* - t_i$	The response time of task T_i
$l = \max(0, c_i - d_i)$	The lateness/tardiness of task T_i

one task that is one hour late (Sagar and Jerald 2022; Qin et al. 2023). Therefore, the total tardiness complements the late task ration metric. We calculate the total tardiness as defined in Eq. 3.

$$T_{sum} = \sum_{i=1}^n l_i [s] \quad (3)$$

DF4: Use real-time capability metric. Computation time of an approach to infer schedules is usually measured to determine real-time capability. We specify DP4 using a real-time capability metric based on the ratio between execution time and computation time of a schedule following a concept mentioned by Ogorelysheva et al. (2023). The real-time capability metric is defined in Eq. 4.

$$E_r = \frac{\sum_{i=1}^n e_i}{C_{max}} \in [0; 1] \quad (4)$$

DF5: Use average response time metric. Complementing the real-time capability metric, the benchmark also needs to consider the response time incorporating the time until an AGV actually starts completing a task after task definition. We calculate the average response time as described in Eq. 5.

$$\hat{H} = \frac{\sum_{i=1}^n h_i}{n} [s] \quad (5)$$

DF6: Provide aggregated score. To facilitate result interpretation for the decision-maker in selecting an AGV fleet scheduling approach (cf. DP2), the benchmark needs to provide an aggregated numeric score as defined in Eq. 6. A single score facilitates comparability among scheduling approaches supporting the decision-maker. The use of an aggregated numeric score is in line with recommendations for benchmark design towards comprehensibility (Bartz-Beielstein et al. 2020) and is also evident in existing benchmarks across various domains (e.g., Coleman et al. 2017; Lin et al. 2014; Stahmann and Rieger 2023).

DF7: Use weights to prioritize metrics. To make the assessment of AGV fleet scheduling approaches independent of the use case, the benchmark enables decision-makers to configure weights for metric prioritization. Equation 6 defines the weighted sum computation of the aggregated score.

$$S_B = w_1 \cdot P - w_2 \cdot T_r - w_3 \cdot T_{sum} - w_4 \cdot E_r - w_5 \cdot \hat{H} \quad (6)$$

Figure 5 shows the benchmarking procedure. First, a set of approaches for AGV fleet scheduling is applied to a problem instance. Second, the decision-maker applies weights to the metrics included in the benchmark (cf. DP1). In the third step, the decision-maker applies the benchmark to the results of the scheduling approaches. Fourth, the benchmark outputs a ranking of the scheduling approaches

so that the decision-maker can select the best approach in step five. The benchmark uses a generic application programming interface (API) to be applicable to various instances of AGV fleets and the results of scheduling approaches.

Notably, we have designed the API to retain broad compatibility across use cases by making few assumptions and providing configurability. See Figure C1 in Online Appendix C for the corresponding UML class diagram. The benchmark's depiction of a schedule and its tasks follows the notation from Table 3, requiring only user input that is expected to be readily available in, and compatible with, typical scheduling contexts. Individual methods implement the computation of the metrics required by DF1-DF5. An additional method computes the aggregated score and allows a user-defined set of weights as input to the score computation, fulfilling DF6 and DF7.

5 Demonstration and Evaluation

5.1 Naturalistic Evaluation: Practitioner Interviews

Naturalistic evaluations investigate how a solution technology functions within its actual context to capture the intricate dynamics of human practices (Venable et al. 2016). In terms of a naturalistic evaluation, we conducted interviews with four practitioners to understand their assessment of our design knowledge (cf. Table 4 for details).

To this end, we selected experts from companies in the intralogistics automation and research industry with relevant experience with both intralogistics and AGVs. We required these experts to have at least five years of relevant experience. Furthermore, we required experts to have roles that include regular work in software implementation, system planning, and management interaction in the context of AGV fleets. Two of our experts were robotics researchers, working on FMS research projects to improve stability and performance in intralogistics automation using AGVs. One expert was a team lead in robotics research, combining the work of a robotics researcher with an increased view from the management perspective. Another expert was a senior expert research & development, advising the development and implementation of AGVs in intralogistics automation. We explained the DPs and asked for their evaluation using a semi-structured questionnaire. Further, we provided an artificial evaluation applying the instantiated benchmark to a real-world scenario and asked the experts for their assessment.

In terms of DP1, all experts approved of throughput, responsiveness, and SLA as three evaluation factors and state that their incorporation in the benchmark would

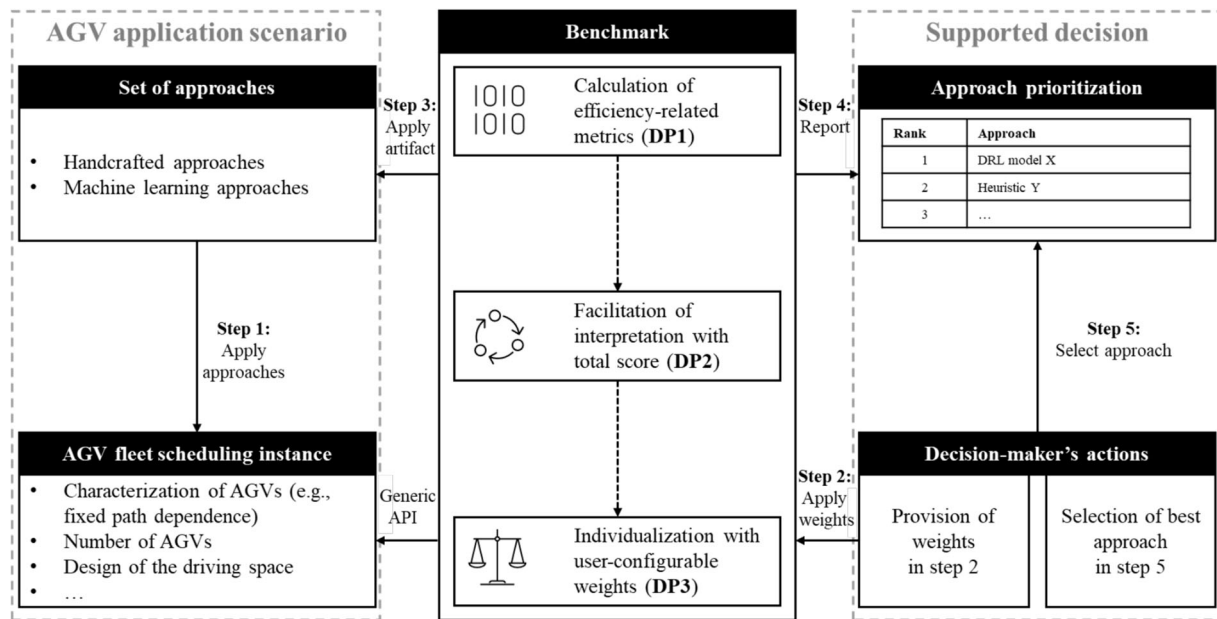


Fig. 5 Design knowledge: design requirements, design principles, and design features

Table 4 Experience of the experts in the fields of AGVs and intralogistics

#	Experience with AGVs	Experience in intralogistics	Role
E6	7 years	7 years	Robotics researcher
E7	14 years	8 years	Senior expert research & development
E8	12.5 years	12.5 years	Team lead robotics research
E9	5 years	5 years	Robotics researcher
Average	9.63 years	8.13 years	

improve the selection decision in practice. Throughput in particular is emphasized by E8 as a relevant metric. In this context, E6 states: “Accordingly, they [the factors] also enable a faster decision [...]. Of course, this also has an influence on how quickly a decision can be made and how well these local problem scenarios can be looked at.” The experts mentioned alternative factors and related metrics that are known from practice. A complete inclusion of all metrics into the benchmark is infeasible, as this depends on the level of knowledge and, in particular, on the needs of the implementation. Thus, the metrics used could be made configurable in some respect to better suit the decision maker and the problem instance (see evaluation of DP3).

The aggregation to a score from DP2 was found to be useful in combination with the possibility of a drill-through to individual metrics. E6 pointed out that the evaluation of approaches using a table showing the different metrics is familiar from practice (cf. Table 5). In comparison, the total score for decision-making was perceived as more valuable. In this vein, E6 argued: “However, if I now imagine that I should make a decision based on the table, then my first overview would still be a bit difficult because there are a lot of metrics that would have to be sorted. And

Table 5 Expert ratings

	E6	E7	E8	E9	Avg.
Accessibility	5	6	6	5	5.5
Importance	4	4	5	3	4
Novelty and insightfulness	4	5	6	4	4.75
Actability and appropriate guidance	6	4	5	5	5
Effectiveness	6	5	5	6	5.5

I find it very helpful to have a general score that combines the individual metrics in order to get a quick start and see: ‘This approach is very good overall’ and then dive deeper into the respective categories in a further step.”

Regarding DP3, the possibility of weighting individual metrics was considered in connection with the statements on DP1. All experts found the option of adapting to local conditions useful. Removing metrics that should not be used for individual implementations was perceived as value-adding. The calibration of the weighting was considered difficult and associated with a high cognitive effort.

In this context, E8 stated: “[...] because I think that you can optimize this entire decision-making process through weighting so that you can definitely achieve a better result for your use case. So, no matter how high the effort is and no matter how modest it is, such a weighting is necessary and makes perfect sense, yes.”

The experts ranked statements on the usability of the DPs in practice on a 6-point Likert scale (1 = total disagreement to 6 = total agreement). Specifically, the statements related to accessibility, importance, novelty and insightfulness, actability and appropriate guidance as well as effectiveness (cf. Iivari et al. 2021). Table 5 shows the results. Notably, the experts rated the statements on all five categories positively, that is, above three (tend to disagree). Accessibility of the targeted result of the benchmarking as well as its effectiveness were rated the highest on average.

5.2 Artificial Evaluation: Prototypical Implementation and Application to Case Study

Artificial evaluation entails a reductionist approach that abstracts from the natural environment to ensure rigorous assessment of the technology artifact’s efficacy. Controlled settings help mitigate obvious confounding variables and reduce the likelihood of misinterpretation and bias (Venable et al. 2016). Against this background, we provide a proprietary implementation of the benchmark to evaluate our design knowledge as outlined in the following.

5.2.1 Description of Problem Instance and Case Study

For the evaluation, we refer to an installed and validated AGV fleet as part of the problem instance that we used to conduct extensive simulations (Roidl 2022). The AGVs were applied in an industrial intralogistics sorting context, as could be found, for example, in warehouses for parcel processing. This use case imposes strong requirements regarding the time to schedule incoming tasks. Sorting systems with traditional belt sorters often achieve throughputs upwards of 1.000 transports/h, with large logistics centers reaching throughputs exceeding 40.000 transports/h (Jodin and Hompel 2012). To achieve such throughputs in an online scheduling context, each individual scheduling decision needs to be computed in a fraction of a second. This AGV fleet combines state-of-the-art technology with novel innovations, constituting a special use case drawing from both traditional and future AGV requirements. This promises a more complete and future-proof evaluation of our benchmark. These novel, highly dynamic AGVs have achieved velocities of up to 12 m/s in space-limited test setups, and are expected to achieve maximum velocities of up to 25 m/s given sufficient space. Depending on the details of the setup, the AGVs achieve an

average velocity of 4–5 m/s. The case study constitutes a realistic AGV fleet scheduling scenario including custom state-of-the-art simulation software, based on the kinematically accurate planning algorithms introduced by Roidl (2022).

In our artificial evaluation case study, a decision-maker sought a scheduling approach for each of the four problem instances to optimize the scheduling of an AGV fleet that transported cargo, moving one item per robot at a time. The problem instance design is a 2×2 factorial design with the factors size (small/large) and alignment (symmetric, asymmetric). AGVs do not have proprietary decision-making capacity, so that the scheduling decisions are *centralized*. The schedules are updated *online* and AGVs are flexible and *do not follow fixed guidance paths*.

In each problem instance, the transportation locations are differentiated as infeeds and outfeeds depending on whether cargo is loaded or off-loaded by an AGV. In the case study, the facility consisted of infeeds, outfeeds, queues, and paths connecting each of these locations as visualized in Fig. 6 (and Fig. D1 in Online Appendix D). Idle AGVs waited in queues at either of the infeeds. A scheduling task was a tuple (i, o) , where cargo needed to be transported from infeed i to outfeed o . Scheduling produced an assigned task $(q_s, p_{q_s i}, p_{i o}, q_t, p_{o q_t})$, where an AGV started from the start queue q_s , drove along paths $p_{q_s i}$ and $p_{i o}$ to collect and deliver the cargo, and then drove to the target queue q_t along path $p_{o q_t}$ to await its next task. Following the core concepts of the AGV fleet as introduced by Roidl (2022), scheduling was strongly intertwined with routing in our use case. While we did not approach both problems with a singular algorithm to solve an integrated scheduling and routing problem, we needed to integrate the routing algorithm introduced by Roidl (2022) into our scheduling approaches to some degree, to enable high quality scheduling decisions. We attempted to solve the scheduling problem $(i, o), s_c \rightarrow (q_s, p_{q_s i}, p_{i o}, q_t, p_{o q_t})$, where s_c was the current system state. This transformed an incoming task into an assigned task, fulfilling the required transport of cargo.

5.2.2 Description of Scheduling Approaches

To provide some diversity to the benchmarks results, we implemented a baseline as well as four dissimilar scheduling approaches, two traditional approaches and two ML-based approaches:

First, we implemented a random approach as a baseline. This approach selects random queues and paths, but ensures that the selection is valid given the current state of the system. It first evaluates which queues are available to use as a start and target, and which paths are available to

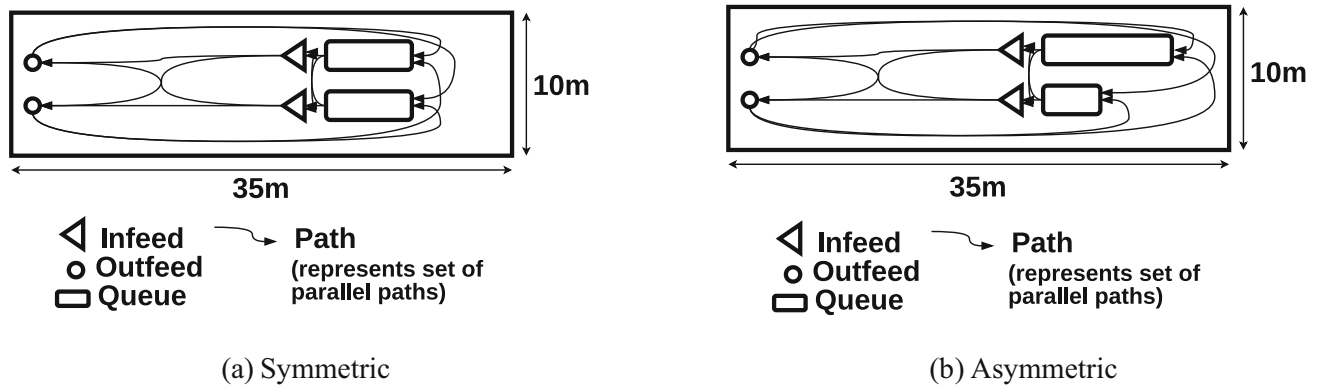


Fig. 6 Small scenario

fulfill the task, following the five components of the assigned task. Then, each component of the assigned task is sampled from a uniform distribution over the available options.

Second, we implemented a traditional heuristic utilizing the shortest travel time first logic (Klei and Kim 1996). This pre-computes the expected travel time for each queue and path, assuming maximum speed traversal without collisions. The fastest option is selected. Forecasting the expected travel times is computationally expensive. We generally expect this approach to yield good results. However, the approach may focus load on few queues and paths that provide a good expected travel time, leading to congestion and lower performance when running the system for longer durations, where it may be beneficial to balance load to queues and paths with a longer expected travel time.

Third, we implemented a traditional round-robin algorithm (Pinedo 2022; Zhang et al. 2026) (denoted R-R in Figures and Tables). This algorithm is a straightforward scheduling approach that guarantees a fair and balanced distribution of the schedule. Regarding our use case, this means equal use of queues and paths. The round-robin algorithm cycles through the set of available queues and paths, skipping unavailable options if necessary. This ensures a steady flow of vehicles for each queue and path, balancing the load. However, this may lead to problems in asymmetric layouts, where a uniform distribution of load may not be desirable.

Lastly, we implemented two ML-based approaches, more specifically two DRL-based approaches. We opted for DRL-based approaches as our representation of the group of ML-based approaches for computation time reasons. As described in previous work by Roidl (2022) and Ogorelysheva et al. (2023), planning the AGVs' trajectories with the required kinematic accuracy to support the high dynamics yields long computation times. There is no straightforward method to predict transport times without

executing the full trajectory computation. Therefore, evaluating different schedules for optimization is expensive. In contrast to other ML-based approaches that are optimized per-instance, DRL-based approaches are optimized in a pre-processing step. This limits most computation expenses to the pre-processing, significantly reducing computation time in production, as inference of the DRL models is fast.

We trained our neural network models to maximize throughput while ensuring stable operation. We opted for two widely adopted DRL models, PPO (Schulman et al. 2017), and soft actor-critic (SAC) (Christodoulou 2019). Both have shown peak performance in a variety of applications (Xu et al. 2021). We modeled our observation and action space following our mathematical problem definition $(i, o), s_c \rightarrow (q_s, p_{q,i}, p_{io}, q_t, p_{oq_t})$. Our observation space includes information on the task, and the current state of AGVs and queues. For the task, we included the index and two-dimensional position of the infeed i and the outfeed o . For the AGVs, we included the two-dimensional position, velocity, and acceleration vectors. For the queues, we included information on using it as a start and target queue. For using a queue as a start queue, we included the current number of AGVs, the time of next AGV availability, and the exit position of the queue. For using a queue as a target queue, we included the capacity, the time of the next possible entry, and the entry position of the queue. We modeled our actions as discrete indices directly following our allocated task definition $(q_s, p_{q,i}, p_{io}, q_t, p_{oq_t})$ by indexing the queues and paths in the scenario.

We used fully connected Multi-Layer Perceptrons (MLPs) as the architecture for our neural networks. Each target function in PPO (policy-function, value-function) and SAC (policy-function, Q-functions, Q-target-functions) was modeled as a separate neural network. Our networks use three common layers with 512 neurons and Rectified Linear Unit (ReLU) activation, and separate heads consisting of another 512 layers with ReLU

activation and an output layer. The output layer for value functions consists of a single output neuron, while the output layers for the action functions consist of output neurons matching the size of the discrete action, with a softmax activation to produce the probabilities for a categorical distribution.

We designed our reward function to maximize long-term throughput by rewarding a steady increase in throughput until a configurable threshold was reached. We provide an in-depth explanation of our DRL-based approaches in Online Appendix C.

5.2.3 Experimental Setup

We conducted experiments on four scenarios by controlled manipulations of the problem instances following Roidl (2022). We implemented a small and a large scenario. For both, we used two variants with the same infeeds and outfeeds, but with different queue capacities: one had symmetric capacities, and the other had asymmetric capacities. Figure 6 depicts the asymmetric and symmetric variants of the small scenario. The paths between the infeed and outfeed pairs each represent a set of three parallel paths, as described by Roidl (2022). We deployed a fleet of 10 AGVs in the small scenario, and 20 AGVs in the large scenario. We set the maximum velocity to 15 m/s.

We trained our DRL-based AGV fleet scheduling models using PPO and SAC, selecting the best model for each. Details of the training process are included in Online Appendix C. Using the benchmark, we collected scheduling data for the selected approaches. We tested different amounts of tasks to evaluate the saturation of calculations. Each experiment ran 20 times, and we reported averages and standard deviations to mitigate the effect of variations due to individual combinations of tasks. The timeliness metrics assume a deadline that is 45 s after the arrival time of each task. Based on observations during development, this is a challenging but realistic to achieve deadline for our simulated AGV fleet.

We selected the benchmark score weights according to our initial observations, the performance goals of our case study, and the recommendations from our experts. To compute the aggregated benchmark score for our case study, we report three values per benchmark weight in Table 6. The scaling factor aims to convert our results to a common order of magnitude. We based this factor on our initial observations of the data. The weight in the second column denotes a metric's importance. This follows the guidance of our experts, where one expert explicitly recommended throughput to be weighted with twice the importance of the other metrics. The final column reports the product of scaling factor and weight, which is the input to the scoring function. We report the results for the small

scenario in detail in Tables 7 and 8. The results for the large scenario are analogous and can be found in Online Appendix D in Tables D1 and D2.

Each Table denotes the results for one scenario, containing the mean and average per combination of metric, scheduling approach, and task count. The best result per metric and task count is highlighted by bold digits.

5.2.4 Experimental Results

Per metric, an increasing task count has a similar effect on all approaches. The throughput increases with increasing task count but starts to converge at approximately 128 tasks. The late task ratio is unaffected at different task counts. The total tardiness increases with larger task counts, as a similar late task ratio with more tasks inevitably leads to higher total tardiness. The real-time capability ratio increases (i.e., computation time is relatively worse) with higher task counts, as more complex collision avoidance needs to be computed during scheduling. The average response time is unaffected by an increasing task count.

We determined the statistical significance of our results using Welch's *t*-test (Welch 1947). With the exception of the late task ratio between the heuristic and the PPO model, and few additional exceptions at low task counts, our results are statistically significant ($\alpha = 0.05$). Overall, the random policy leads to the worst results in all metrics across all experiments, except for the real-time capability metric. As making random scheduling decisions is fast, and an inefficient schedule leads to a long makespan, the ratio between computation time and execution time is small.

Throughput is generally around 2% higher in the symmetric scenario compared to the asymmetric scenario. In the symmetric scenario, the round-robin algorithm achieves the best results, performing around 0.1% better compared to the heuristic, 1.1% better than the PPO model and 7% better than the SAC model in the small symmetric scenario. In the asymmetric scenario, the PPO model achieves the best results, performing around 1% better than the heuristic, 9.5% better than the SAC model, and 9.7% better than the round-robin algorithm. The late task ratio fluctuates at around 1% for the heuristic, round-robin algorithm and PPO model in the small scenario. The SAC model performs around 50–100% worse. In the asymmetric scenario, the heuristic and PPO model again fluctuate at a ratio of around 1%, with the heuristic slightly outperforming the PPO model. The SAC policy performs around 200% worse, and the round-robin algorithm performs around 300% worse. The best result in total tardiness fluctuates between the heuristic, the round-robin algorithm, and the PPO model. Especially in the symmetric scenario, there are no clear trends, and the standard deviation is high compared to the

Table 6 Benchmark score weights for the case study

Metric	Unit	Value range	Scaling	Weight	Final weight
Throughput	transports/h	ca. [400; 1500]	1	2	$w_1 = 2$
Total delay	seconds	[0; 1000]	1	1	$w_2 = 1$
Delay ratio	(ratio)	[0; 1]	500	1	$w_3 = 500$
Computation time ratio	(ratio)	ca. [0; 1]	500	1	$w_4 = 500$
Response time	seconds	ca. [0; 10]	50	1	$w_5 = 50$

Table 7 Experiment results small symmetric scenario

Metric	Approach	32 Tasks	64 Tasks	128 Tasks	256 Tasks	512 Tasks
Throughput	Random	706.48 $\pm$ 40.95	755.00 $\pm$ 23.23	779.37 $\pm$ 37.58	787.48 $\pm$ 24.69	801.62 $\pm$ 17.29
	Heuristic	834.53 $\pm$ 16.71	904.90 $\pm$ 13.90	952.21 $\pm$ 8.74	975.15 $\pm$ 6.02	983.33 $\pm$ 3.91
	R-R	839.18 $\pm$ 12.86	905.68 $\pm$ 14.56	949.73 $\pm$ 8.27	974.62 $\pm$ 5.87	985.79 $\pm$ 3.33
	PPO	826.92 $\pm$ 17.84	898.35 $\pm$ 15.89	940.36 $\pm$ 9.16	960.81 $\pm$ 5.91	972.97 $\pm$ 6.00
	SAC	783.94 $\pm$ 17.11	842.44 $\pm$ 22.58	888.41 $\pm$ 13.11	911.14 $\pm$ 9.06	917.78 $\pm$ 6.96
Late task ratio	Random	0.036 $\pm$ 0.035	0.035 $\pm$ 0.015	0.047 $\pm$ 0.018	0.050 $\pm$ 0.013	0.049 $\pm$ 0.009
	Heuristic	0.011 $\pm$ 0.015	0.009 $\pm$ 0.011	0.009 $\pm$ 0.008	0.010 $\pm$ 0.006	0.009 $\pm$ 0.003
	R-R	0.006 $\pm$ 0.013	0.007 $\pm$ 0.010	0.007 $\pm$ 0.007	0.011 $\pm$ 0.006	0.011 $\pm$ 0.004
	PPO	0.014 $\pm$ 0.016	0.008 $\pm$ 0.010	0.009 $\pm$ 0.010	0.010 $\pm$ 0.005	0.011 $\pm$ 0.005
	SAC	0.022 $\pm$ 0.017	0.018 $\pm$ 0.013	0.013 $\pm$ 0.009	0.017 $\pm$ 0.009	0.018 $\pm$ 0.004
Total tardiness	Random	9.85 $\pm$ 14.08	14.43 $\pm$ 8.18	45.83 $\pm$ 19.99	91.24 $\pm$ 35.90	171.96 $\pm$ 65.24
	Heuristic	1.46 $\pm$ 2.76	3.74 $\pm$ 5.47	6.65 $\pm$ 6.79	11.18 $\pm$ 9.95	23.30 $\pm$ 16.49
	R-R	1.32 $\pm$ 2.99	2.07 $\pm$ 3.30	2.58 $\pm$ 4.50	19.58 $\pm$ 11.83	28.64 $\pm$ 19.43
	PPO	2.33 $\pm$ 4.41	1.61 $\pm$ 2.39	8.14 $\pm$ 10.56	15.98 $\pm$ 14.14	27.58 $\pm$ 18.74
	SAC	3.84 $\pm$ 4.51	6.31 $\pm$ 7.14	10.12 $\pm$ 9.55	27.35 $\pm$ 19.96	54.20 $\pm$ 22.86
Real-time capability	Random	0.159 $\pm$ 0.017	0.182 $\pm$ 0.008	0.196 $\pm$ 0.009	0.194 $\pm$ 0.008	0.200 $\pm$ 0.004
	Heuristic	0.265 $\pm$ 0.006	0.293 $\pm$ 0.005	0.308 $\pm$ 0.004	0.317 $\pm$ 0.002	0.321 $\pm$ 0.002
	R-R	0.165 $\pm$ 0.003	0.187 $\pm$ 0.004	0.198 $\pm$ 0.003	0.205 $\pm$ 0.002	0.208 $\pm$ 0.001
	PPO	0.170 $\pm$ 0.004	0.191 $\pm$ 0.003	0.203 $\pm$ 0.004	0.210 $\pm$ 0.002	0.214 $\pm$ 0.002
	SAC	0.174 $\pm$ 0.006	0.202 $\pm$ 0.006	0.217 $\pm$ 0.003	0.224 $\pm$ 0.003	0.228 $\pm$ 0.002
Average response time	Random	9.38 $\pm$ 0.76	9.43 $\pm$ 0.35	9.62 $\pm$ 0.45	9.71 $\pm$ 0.26	9.66 $\pm$ 0.23
	Heuristic	7.18 $\pm$ 0.25	7.27 $\pm$ 0.13	7.22 $\pm$ 0.07	7.21 $\pm$ 0.06	7.24 $\pm$ 0.03
	R-R	7.22 $\pm$ 0.15	7.27 $\pm$ 0.16	7.24 $\pm$ 0.09	7.23 $\pm$ 0.05	7.22 $\pm$ 0.03
	PPO	7.33 $\pm$ 0.15	7.37 $\pm$ 0.14	7.33 $\pm$ 0.06	7.35 $\pm$ 0.05	7.34 $\pm$ 0.05
	SAC	7.97 $\pm$ 0.32	8.05 $\pm$ 0.25	7.94 $\pm$ 0.16	7.95 $\pm$ 0.08	7.99 $\pm$ 0.07
Score (unweighted)	Random	687.05 $\pm$ 49.84	730.93 $\pm$ 26.96	723.68 $\pm$ 51.43	686.29 $\pm$ 54.92	619.75 $\pm$ 79.63
	Heuristic	825.61 $\pm$ 17.98	893.59 $\pm$ 15.78	938.03 $\pm$ 13.04	956.44 $\pm$ 12.65	952.46 $\pm$ 15.24
	R-R	830.47 $\pm$ 12.64	896.15 $\pm$ 15.13	939.71 $\pm$ 7.14	947.60 $\pm$ 13.47	949.72 $\pm$ 19.47
	PPO	817.08 $\pm$ 18.81	889.18 $\pm$ 15.60	924.67 $\pm$ 13.52	937.26 $\pm$ 16.14	937.83 $\pm$ 19.70
	SAC	771.94 $\pm$ 18.50	827.85 $\pm$ 25.84	870.12 $\pm$ 15.15	875.61 $\pm$ 24.56	855.34 $\pm$ 26.54
Score (weighted)	Random	836.67 $\pm$ 121.99	915.56 $\pm$ 60.45	910.37 $\pm$ 110.90	876.49 $\pm$ 90.71	824.22 $\pm$ 106.66
	Heuristic	1170.60 $\pm$ 43.55	1291.04 $\pm$ 34.38	1378.06 $\pm$ 24.37	1415.22 $\pm$ 20.84	1416.31 $\pm$ 15.84
	R-R	1230.80 $\pm$ 28.45	1348.61 $\pm$ 36.18	1432.55 $\pm$ 17.70	1460.22 $\pm$ 20.60	1472.90 $\pm$ 21.46
	PPO	1192.99 $\pm$ 37.57	1327.23 $\pm$ 37.20	1400.08 $\pm$ 25.83	1428.37 $\pm$ 22.96	1439.22 $\pm$ 24.89
	SAC	1067.93 $\pm$ 50.63	1165.96 $\pm$ 58.28	1254.60 $\pm$ 33.28	1277.28 $\pm$ 35.08	1259.07 $\pm$ 34.52

mean values. In the asymmetric scenario, the heuristic and PPO model perform significantly better compared to the

round-robin algorithm and SAC model, achieving a total tardiness that is around two to four times lower,

Table 8 Experiment results small asymmetric scenario

Metric	Approach	32 Tasks	64 Tasks	128 Tasks	256 Tasks	512 Tasks
Throughput	Random	569.15 $\pm$ 100.88	647.63 $\pm$ 62.92	591.55 $\pm$ 77.00	577.86 $\pm$ 52.30	580.45 $\pm$ 33.33
	Heuristic	816.21 $\pm$ 31.33	886.18 $\pm$ 20.84	917.84 $\pm$ 15.64	939.21 $\pm$ 13.83	947.33 $\pm$ 9.57
	R-R	761.35 $\pm$ 10.93	811.15 $\pm$ 8.25	843.13 $\pm$ 5.64	861.14 $\pm$ 3.64	870.66 $\pm$ 2.82
	PPO	816.66 $\pm$ 12.80	890.16 $\pm$ 10.26	935.15 $\pm$ 6.83	950.10 $\pm$ 5.05	963.62 $\pm$ 4.81
	SAC	765.93 $\pm$ 25.00	820.81 $\pm$ 32.24	844.37 $\pm$ 13.22	855.41 $\pm$ 18.19	870.52 $\pm$ 14.14
Late task ratio	Random	0.131 $\pm$ 0.077	0.097 $\pm$ 0.046	0.154 $\pm$ 0.054	0.167 $\pm$ 0.035	0.166 $\pm$ 0.029
	Heuristic	0.006 $\pm$ 0.016	0.011 $\pm$ 0.015	0.014 $\pm$ 0.009	0.009 $\pm$ 0.006	0.009 $\pm$ 0.004
	R-R	0.027 $\pm$ 0.023	0.038 $\pm$ 0.015	0.038 $\pm$ 0.014	0.046 $\pm$ 0.017	0.043 $\pm$ 0.009
	PPO	0.009 $\pm$ 0.017	0.011 $\pm$ 0.011	0.014 $\pm$ 0.010	0.010 $\pm$ 0.005	0.012 $\pm$ 0.004
	SAC	0.017 $\pm$ 0.023	0.026 $\pm$ 0.020	0.029 $\pm$ 0.016	0.027 $\pm$ 0.011	0.027 $\pm$ 0.006
Total tardiness	Random	48.30 $\pm$ 39.47	63.90 $\pm$ 39.77	260.11 $\pm$ 137.18	583.77 $\pm$ 201.49	1177.72 $\pm$ 232.18
	Heuristic	1.11 $\pm$ 2.65	3.79 $\pm$ 5.70	9.66 $\pm$ 8.87	17.57 $\pm$ 22.24	27.55 $\pm$ 15.63
	R-R	4.59 $\pm$ 6.42	16.08 $\pm$ 11.43	30.04 $\pm$ 16.87	74.91 $\pm$ 29.77	130.99 $\pm$ 35.56
	PPO	0.98 $\pm$ 2.21	4.01 $\pm$ 4.86	10.64 $\pm$ 8.79	17.54 $\pm$ 11.57	29.65 $\pm$ 15.85
	SAC	2.15 $\pm$ 2.95	13.50 $\pm$ 12.43	24.53 $\pm$ 18.86	43.60 $\pm$ 22.77	86.53 $\pm$ 32.94
Real-time capability	Random	0.114 $\pm$ 0.028	0.141 $\pm$ 0.014	0.121 $\pm$ 0.016	0.116 $\pm$ 0.011	0.117 $\pm$ 0.008
	Heuristic	0.239 $\pm$ 0.010	0.266 $\pm$ 0.007	0.277 $\pm$ 0.005	0.285 $\pm$ 0.004	0.289 $\pm$ 0.003
	R-R	0.145 $\pm$ 0.006	0.162 $\pm$ 0.003	0.169 $\pm$ 0.002	0.176 $\pm$ 0.002	0.177 $\pm$ 0.002
	PPO	0.169 $\pm$ 0.005	0.194 $\pm$ 0.003	0.203 $\pm$ 0.003	0.211 $\pm$ 0.003	0.216 $\pm$ 0.001
	SAC	0.179 $\pm$ 0.007	0.203 $\pm$ 0.009	0.218 $\pm$ 0.005	0.224 $\pm$ 0.005	0.228 $\pm$ 0.004
Average response time	Random	12.25 $\pm$ 2.61	11.31 $\pm$ 1.19	13.31 $\pm$ 1.90	13.78 $\pm$ 1.34	13.78 $\pm$ 0.90
	Heuristic	7.52 $\pm$ 0.34	7.41 $\pm$ 0.24	7.52 $\pm$ 0.13	7.51 $\pm$ 0.10	7.52 $\pm$ 0.08
	R-R	8.64 $\pm$ 0.32	9.29 $\pm$ 0.31	9.61 $\pm$ 0.39	9.77 $\pm$ 0.28	9.81 $\pm$ 0.24
	PPO	7.46 $\pm$ 0.22	7.40 $\pm$ 0.16	7.40 $\pm$ 0.06	7.44 $\pm$ 0.04	7.42 $\pm$ 0.04
	SAC	8.28 $\pm$ 0.42	8.50 $\pm$ 0.45	8.65 $\pm$ 0.25	8.66 $\pm$ 0.24	8.62 $\pm$ 0.14
Score (unweighted)	Random	508.36 $\pm$ 137.84	572.19 $\pm$ 97.64	317.86 $\pm$ 213.41	-19.97 $\pm$ 253.71	-611.33 $\pm$ 264.62
	Heuristic	807.34 $\pm$ 31.31	874.69 $\pm$ 22.06	900.37 $\pm$ 18.80	913.84 $\pm$ 24.79	911.96 $\pm$ 15.79
	R-R	747.94 $\pm$ 11.86	785.58 $\pm$ 16.04	803.27 $\pm$ 18.96	776.25 $\pm$ 29.81	729.63 $\pm$ 36.08
	PPO	808.05 $\pm$ 13.47	878.54 $\pm$ 10.87	916.90 $\pm$ 10.00	924.89 $\pm$ 11.84	926.33 $\pm$ 17.06
	SAC	755.31 $\pm$ 26.27	798.58 $\pm$ 41.20	810.94 $\pm$ 25.07	802.91 $\pm$ 34.67	775.11 $\pm$ 43.43
Score (weighted)	Random	354.97 $\pm$ 385.45	547.18 $\pm$ 228.18	119.83 $\pm$ 398.95	-258.37 $\pm$ 380.46	-847.25 $\pm$ 348.73
	Heuristic	1132.83 $\pm$ 73.72	1259.62 $\pm$ 50.92	1304.91 $\pm$ 37.76	1338.17 $\pm$ 36.15	1341.81 $\pm$ 23.22
	R-R	999.94 $\pm$ 32.42	1041.28 $\pm$ 33.34	1072.51 $\pm$ 38.68	1048.31 $\pm$ 44.35	1009.67 $\pm$ 45.06
	PPO	1170.20 $\pm$ 37.49	1304.20 $\pm$ 23.66	1381.45 $\pm$ 18.23	1400.16 $\pm$ 15.54	1412.51 $\pm$ 22.41
	SAC	1018.07 $\pm$ 70.99	1088.55 $\pm$ 96.14	1108.12 $\pm$ 50.43	1109.07 $\pm$ 62.15	1095.86 $\pm$ 61.78

respectively. While the random approach performs best concerning real-time capability, the round-robin algorithm, PPO model and SAC model also achieve competitive results. The real-time capability of the heuristic is especially concerning in the large symmetric scenario, where it exceeds a value of 1 at 256 and 512 tasks, indicating that the approach is not capable of computing in real-time in practice. The average response time behaves similarly to the throughput, where the heuristic and the round-robin algorithm perform best in the symmetric scenario, outperforming the PPO model by about 2%, and the PPO model

outperforming the heuristic by about 1.5% in the asymmetric scenario. The SAC model performs 10 to 15% worse, respectively. The round-robin algorithm performs around 25% worse in the asymmetric scenario.

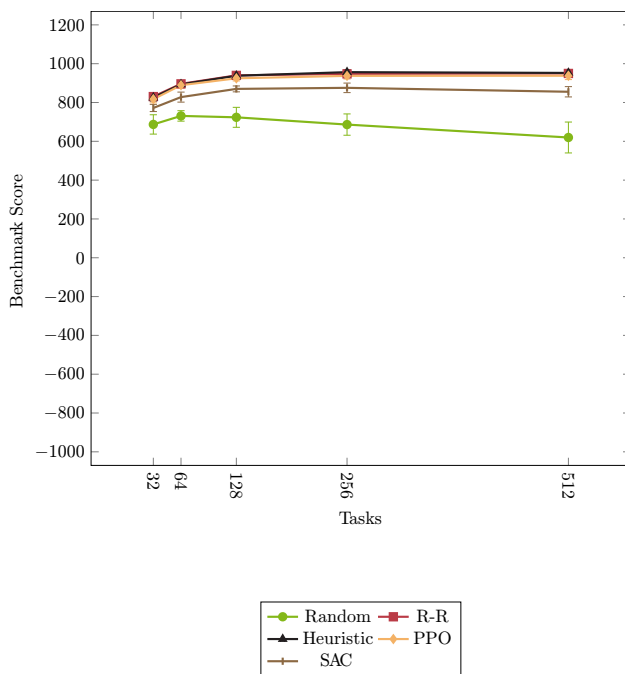
The benchmark score converges with increasing task count. The random approach achieves the lowest score in both the symmetric and asymmetric scenario. All other approaches perform significantly better than the random approach. In the symmetric scenario, the round-robin algorithm performs best, outperforming the PPO model by 2.3%, the heuristic by 4.1%, and the SAC model by 15.3%.

The round-robin algorithm achieved some of the best results across all metrics, yielding its high benchmark score. While the heuristic achieves higher throughput compared to the PPO model, the improved real-time capability of the PPO model leads to its better performance regarding the benchmark score. In the asymmetric scenario, the PPO model performs best, outperforming the heuristic by 4.5%, the SAC model by 22.9%, and the round-robin algorithm by 28.9%. The high benchmark scores stem from the PPO models' good performance across all metrics in the asymmetric scenario.

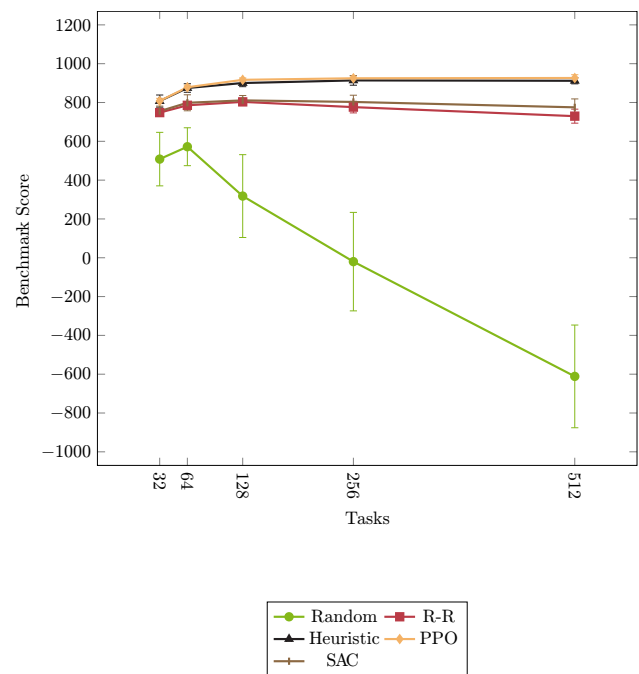
Figures 7 and 8 visualize the results of the aggregated benchmark score, combining the results of the individual metrics. To demonstrate the effect of user-defined weights, Fig. 7 shows the unweighted benchmark score, that is an aggregated sum where all weights are set to one, while Fig. 8 includes user-specified weights. There are two clear differences. First, the range of the values is different. Second, in the symmetric scenario, the unweighted score of the heuristic is higher compared to the unweighted score of the round-robin algorithm and PPO model, while its weighted score is lower than both. This stems from the heuristic's good performance concerning throughput and bad performance concerning real-time capability. As the real-time capability is captured as a ratio, usually measuring between zero and one, its effect on the unweighted score is insignificant compared to the throughput which measures at around 1000. The weighting mitigates this

difference in order of magnitude, allowing the real-time capability to have the desired effect on the overall score. In general, the results in the large scenario confirm the ones we found in the small scenario.

Figure 8 also demonstrates the advantages of the aggregated score (DF6) facilitating result interpretation (DP3). The benchmark not only incorporates the evaluation metrics for prioritizing approaches, derived from both literature and practice, but also aggregates these into a comprehensive score. This score considers the individual weights assigned by decision-makers, facilitating a transparent assessment of which approach yields the optimal results across different numbers of runs. In contrast to Tables 7 and 8 (as well as Tables D1 and D2 in Online Appendix D), which present metrics in isolation, the figures of the scores provide a more integrated view. They transparently indicate that the PPO model is most suitable for scheduling this AGV fleet. The figures also show that in the symmetric case, the PPO model and the heuristic have almost similar suitability. Using the PPO model would support the decision-maker's confidence in the selected scheduling approach, as it aligns most closely with the weighted preferences among all evaluated approaches. Further, as it scores better in asymmetric scenarios it would render the AGV fleet more robust in case of environment changes from symmetric to asymmetric layouts.



(a) Symmetric



(b) Asymmetric

Fig. 7 Small scenario: unweighted score

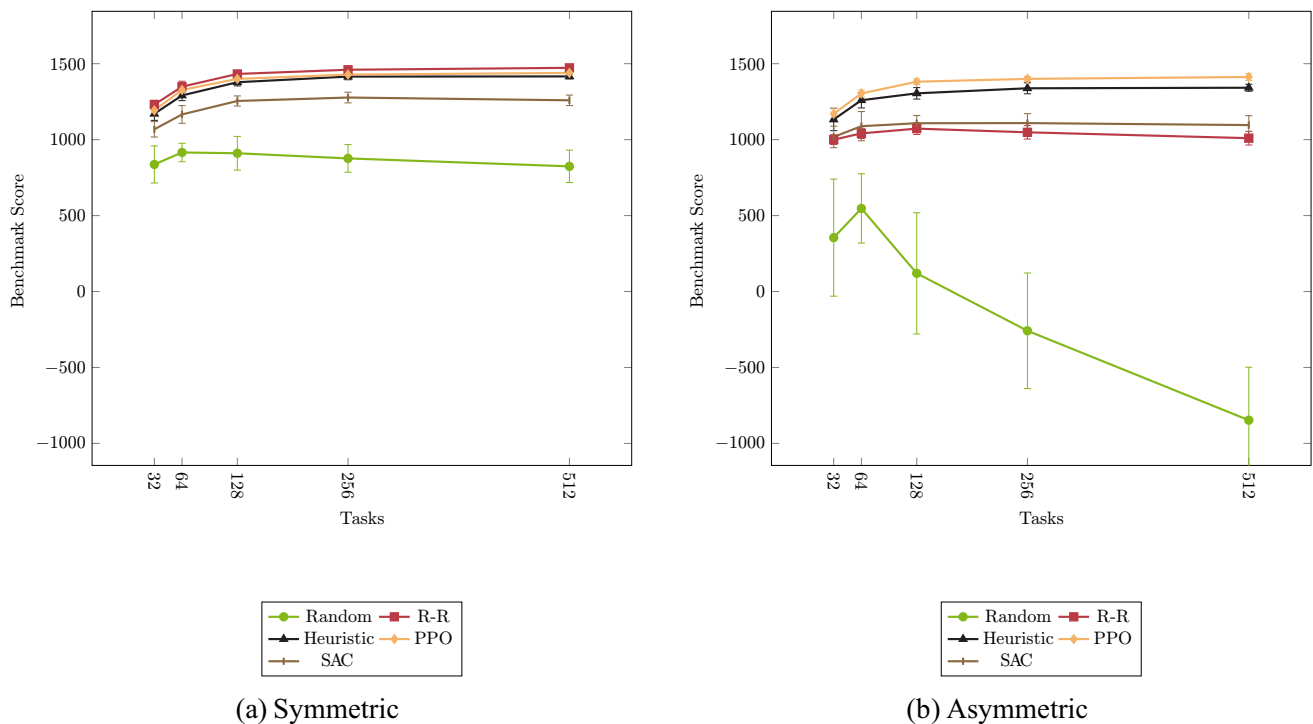


Fig. 8 Small scenario: weighted score

6 Discussion, Implications, and Limitations

6.1 Discussion

The efficiency-optimized scheduling of AGVs is a complex problem, particularly considering real-time updates of schedules and non-fixed guidance paths. In recent years, the number of scheduling approaches has proliferated significantly, especially with the advent of AI-based methods. Specifically, ML-based and DRL-based approaches have extended the pool of alternative approaches.

Due to increasing interest in practice and academia, several researchers have contributed to clarity in the complex field of AGV scheduling, often through structured literature reviews (e.g., Oyekanlu et al. 2020). We complement this by specifically addressing the increasing diversity of approaches and the lack of decision support for their selection. Our design knowledge targets the use-case-specific prioritization of approaches based on problem-inherent criteria, encapsulated as prescriptive guidance in DPs and DFs.

In this vein, we contribute with design knowledge that can guide researchers as well as practitioners to be more conscious about the parameters in the selection of approaches. As our experiment underlines, there is no best-in-class approach for all scenarios not even on an individual metric. While one approach may excel in symmetric scenarios, it may be less appropriate in asymmetric setups.

This confirms the “No Free Lunch Theorem” for our case and leads to a number of design trade-offs one can identify when using our benchmark design.

The trade-offs do not only manifest when evaluating scenarios, they also manifest when designing the benchmark in the first place. Two trade-offs arise from the design of the DFs in relation to the DSSDRs, which form the basis of the benchmark. The first trade-off involves all three DSSDRs. Fewer assumptions about relevant metrics and the use of more complex scoring functions, such as those supporting non-linear scaling, could enhance decision quality. This would also reduce system restrictions by offering users greater flexibility to adapt the benchmark to their specific use case. However, this is in direct conflict with reducing the cognitive effort, as the user would then need to make significantly more complex decisions about the metrics, weights, and scoring function to use in the benchmark. Therefore, we decided to focus on the most frequent metrics and use a straightforward weighted sum, which only requires the user to adjust five weights.

The second trade-off concerns DSSDR2 and DSSDR3. The benchmark’s API needs to make assumptions about the required data and therefore the supported scheduling models. The user needs to be able to provide the required data and integrate the benchmark with their fleet management system with low cognitive effort to fulfill DSSDR2. Otherwise, a high cognitive load for deployment and implementation could, for example, induce an aversion to

using the benchmark and impair the human decision-making performance.

6.2 Theoretical and Practical Implications

Theoretical implications of our prescriptive design knowledge and our experimental results contribute to the knowledge base of AGV scheduling approach selection by providing guidelines for benchmark establishment, for handling the arising complex decision situations as well as for approach development.

Against the background of proliferating AI-based models for various application domains, we argue for incorporating approach selection research into socio-technical research. In this regard, we extend the knowledge base specifically for AGV fleet scheduling. The inferred DPs and DFs provide normative guidelines that can be adapted to other problem characteristics, such as decentralized scheduling as usual for AMRs. That is, from a theoretical perspective, the results present nascent design knowledge for the use case of AGV scheduling (Gregor and Hevner 2013). The presented design knowledge is formalized to be applicable across application cases, which enables further refinement by researchers to more mature design theories.

Our results also entail implications referring to handling the “No Free Lunch Theorem” (Wolpert and Macready 1997) with DSSs and future approach development. That is, we amplify that there is no “one size fits all” approach in AGV fleet scheduling as it is often the case in complex decision support cases (see for example Heinrich et al. 2021 for the case of predictive process monitoring), we infer design knowledge on how to determine the fit between a variety of approaches and a problem instance. We show that decision support can be designed in accordance with the theorem and aid decision-makers’ solution finding. In this regard, our work integrates into a larger body of research that focuses approach selection in a variety of fields (e.g., Mantovani et al. 2019; Abdulrahman et al. 2019; Wang et al. 2019).

Further, our design knowledge constitutes a foundation for approach development. The benchmark artifact can be used for benchmarking challenges to optimize the use and design of approaches for AGV fleet scheduling. This may also include evaluation of own approaches as well as parametrization of new implementations. Further, the benchmark enables the exploration of priorities through weighting, allowing future scheduling approaches to be directly optimized for these priority settings.

Practical relevance signifies the possibility of proprietary implementations in business practice, of increased trust in schedules for AGV fleets, and of the incorporation of autonomy to decision support in intralogistics.

Our benchmark can be considered a set of smaller, modular artifacts, which are the formalized and implemented ranking criteria. These are flexibly combinable, which allows for customization and amplification using proprietary implementations (cf. vom Brocke et al. 2020). Future implementations can therefore make use of systematic transparency based on performance criteria. Enhancing the possibility to set and test priorities in the criteria, we introduced weights. The modularity allows for customized versions to solve related problems. An amplification could include further metrics with lower levels of diffusion, but potential relevance to single problem instances, such as machine utilization (e.g., Durst et al. 2023; Li et al. 2025b).

The inferred design knowledge contributes to transparency in the dynamic field of AGV fleet scheduling, where a growing number of scheduling approaches hinders selection of suitable approaches. The benchmark artifact provides guidance for approach selection and thus potentially increases trust that the selected approach is best suited for efficient AGV scheduling. Related research, specifically on the use of AI models, shows the pivotal role of trust for decision-makers to accept and work with decisions recommended by AI-based DSSs (Glikson and Woolley 2020). This is especially the case for intransparent AI models with obscured decision logic, such as the models the research considers eligible for the scheduling of AGV fleets. An increase of decision-makers’ trust can reduce aversion against the scheduling approaches (Dietvorst et al. 2015). With a potential increase in trust towards scheduling approaches for AGV fleets, the benchmark artifact can therefore contribute to efficiency in intralogistics based on the selection of most suitable scheduling approaches.

In this vein, we contribute to the ongoing discourse of how to incorporate AI-based autonomy in practical business operations (e.g., Mayr et al. 2024; Adam et al. 2024; blinded). Using our design knowledge, human decision-makers can streamline efforts of trial-and-error and automate the selection of scheduling approaches. Thereby, this work lays the foundation for increasing autonomy in intralogistics practice. Future research can build on this by adapting and transferring design knowledge to AMR, applying a different decision-making focus, where individual vehicles autonomously determine the best routes in a decentralized manner (Fragapane et al. 2021).

6.3 Limitations and Future Work

Our research has limitations in terms of the employed research approach and the results. The derived design knowledge builds on a structured literature review and expert interviews. Regarding the former, the design of the search string and choice of databases has implications on

the results. While we used a broad search string and a variety of databases, a different choice might lead to different results. In the same vein, literature selection and reflection are subjective to the researchers' interpretation. However, we adhered to established guidelines (vom Brocke et al. 2015). Further, we interviewed a limited number of experts, future research might target a larger and more diverse interview study to infer additional results.

The scope for the design of the benchmark is limited to performance optimization. However, practical implementations may be subject to further objectives. For example, interviewed experts mentioned their interest in considering resource utilization or energy consumption. Also, the benchmark score is only comparable given the same preconditions, making it context-sensitive. The preconditions include the environment of the AGV fleet, its size, the number of tasks that need to be scheduled, and the benchmark's weights. Therefore, our results build a foundation for future work to include more different scheduling approaches in different contexts and investigate scoring results for further refinement of the benchmark.

Further, we did not implement a meta-heuristic or stochastic optimization. While these are prevalent in literature, they are not feasibly applicable to our use case. The strict intertwinement between scheduling and trajectory computation severely impacts required computation time. Online, or per-instance, optimization, which explores the search space from scratch for each new transportation task, would imply a significant requirement regarding computation time. As our scheduling problem is an online problem, where tasks arrive one at a time and a scheduling decision needs to be made ad hoc for each task, it is infeasible to use online optimization methods. Therefore, our choice only includes scheduling approaches that may require expensive pre-processing, but are relatively computationally fast at runtime.

While our goal was to evaluate the boundaries of our benchmark including novel innovations in the field of AGVs, we recognize potential for further evaluation. For example, a more traditional use case, for example with less dynamic AGVs using grid-based routing, may be more well-suited for applying additional standard AGV scheduling algorithms from literature, such as meta-heuristics.

Lastly and forward-looking, during several expert interviews, the idea of broadening the benchmark to a more general DSS that could be linked to individual use case scenarios was discussed. The more general DSS could support users in identifying the required customization. Most importantly, such a DSS could help select the appropriate metrics to assess model performance and set the weights to combine these metrics into the benchmark score.

7 Conclusion

Automation of transport in intralogistics is of increasing interest in practice and academia. AGV fleets show high potential to leverage efficiency as they are faster and more flexible than conventional methods. A growing variety of traditional approaches and wML-based approaches has been developed to schedule AGV fleets. Drawing on the "No Free Lunch Theorem", we pose that one cannot know a priori which approach works best for scheduling a specific AGV fleet. To solve this problem, we inferred design knowledge for a benchmark that constitutes a DSS by facilitating approach selection to decision-makers (Storey et al. 2024).

With our research we contribute twofold: First, we derived design knowledge for a generic benchmark capable of ranking approaches for AGV fleet scheduling. The design knowledge was generalized in terms of a formulation of own DRs, DPs, and DFs. Second, we provide a technical instantiation of the benchmark that was developed and evaluated in terms of technical feasibility and practical usefulness. These contributions allow for adaptation and further refinement of the benchmark to individual requirements in theory and practice.

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